

Evolving a real bacterial genome with ZOMBI2’s nucleotide model

The *Escherichia coli* K-12 chromosome along a simulated 12-species tree

ZOMBI2 — nucleotide genic model

Abstract

We take the real annotation of *Escherichia coli* K-12 MG1655 — a single circular chromosome of 4,641,652 bp carrying 4480 genes — as the ancestral genome, and evolve it forward along a simulated 12-species tree with ZOMBI2’s *nucleotide genic* model. Genes are never broken by structural events; losses may pseudogenise a gene instead of deleting it, and transfers may replace a recipient’s homologous locus. From one run we recover a full pan-genome: 4203 core and 294 accessory gene families, 441 families with paralogs, and a reconstructed gene tree for every gene and intergene. This is a worked demonstration that the model scales to a megabase-sized real genome and reproduces textbook bacterial genome dynamics.

1 Model and set-up

The ancestral genome. We read the RefSeq GFF3 of *E. coli* K-12 MG1655 (assembly GCF_000005845.2, chromosome NC_000913.3). ZOMBI2 copies the chromosome length (4,641,652 bp, circular) and the coordinates of its genes; the intergenic regions are simply the gaps between them. Bacterial genes overlap (shared start/stop codons, nested ORFs), and the genic model forbids event break-points *inside* a gene, so overlaps are removed by trimming: of 4506 annotated genes, 768 had their start clipped to the previous gene’s end and 26 nested genes were dropped, leaving 4480 disjoint genes covering 86.8% of the chromosome.

The species tree. A 12-tip tree (Fig. 1) was simulated under a birth–death process ($\lambda = 1.0$, $\mu = 0.35$, crown age 1; seed 42).

The evolutionary model. Genes and intergenes evolve by variable-length structural events whose rates are *per nucleotide*: inversion 4e-06, loss 7e-06, duplication 3e-06, transfer 3e-06 and transposition 2e-06 per site, plus origination 2 per branch. Event lengths are geometric with mean 2,500 bp (extension 0.9996). A loss that hits a gene pseudogenises it — keeping the sequence but marking it non-functional — with probability 0.4; a transfer is a homologous replacement with probability 0.4 (seed 7).

Because genes are never cut, each of the 4480 genes stays a single indivisible unit (an *atom*) wherever it survives, while intergenic DNA fragments freely. The run yields 8358 atoms in total: 4497 gene atoms (the 4480 ancestral genes plus 18 originated *de novo*) and 3861 intergene atoms.

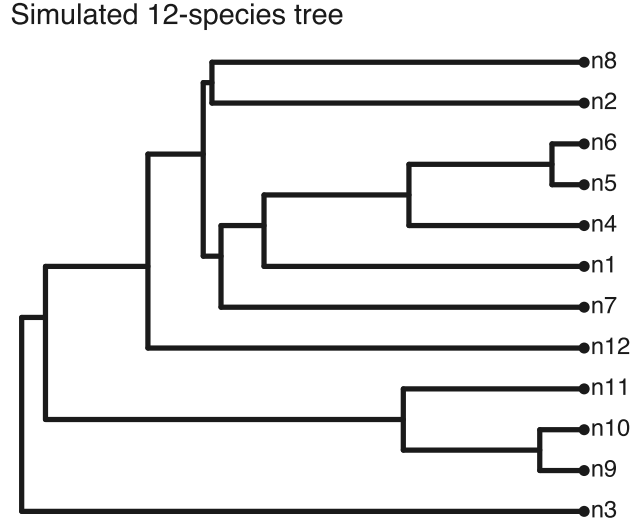


Figure 1. The simulated 12-species tree the genome was evolved along.

2 Results

2.1 Genome dynamics along the tree

Every lineage erodes and innovates (Fig. 2). Averaged over the 12 extant genomes, 98.1 % of the ancestral genes are retained (mean 4395 genes; range 4359–4429) at a mean genome size of 4.674 Mbp, with 50 pseudogenes per genome. Gene loss (red) and pseudogenisation dominate the erosion side; gene duplication (blue) and the occasional *de novo* origination (green) drive innovation.

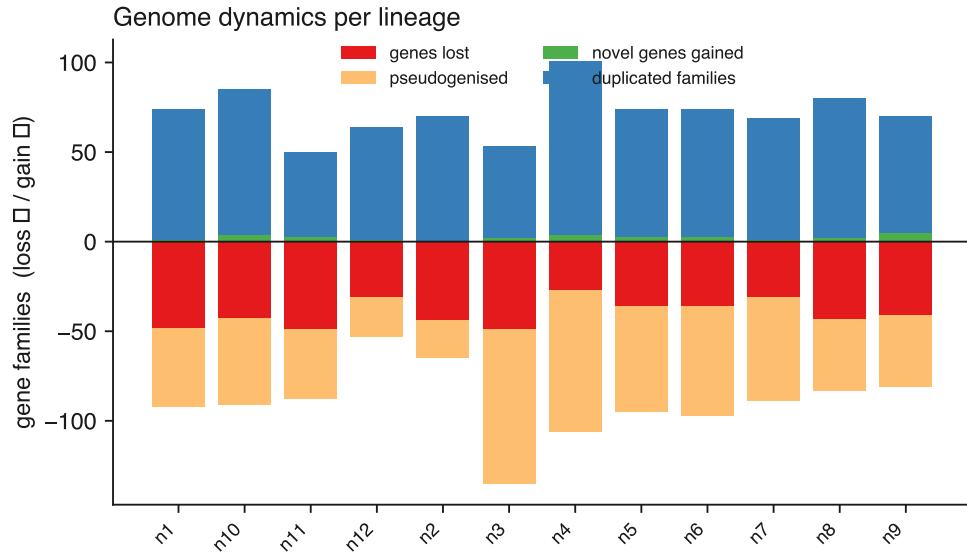


Figure 2. Per-lineage genome dynamics relative to the ancestral *E. coli* gene set: genes lost and pseudo-genised (below the axis) versus novel genes gained and families duplicated (above). Each lineage has its own history of gain and loss.

2.2 A pan-genome emerges

Reading gene presence across the 12 genomes gives the classic bacterial pan-genome (Fig. 3, left): a large **core** of 4203 families present in every species, a long **accessory** tail of 294 families (10 found in a single genome), produced purely by lineage-specific loss, transfer and origination. 441 families carry paralogs in at least one genome. The copy-number matrix of the most variable families (Fig. 3, right) shows the gains (dark), single copies (salmon) and losses (pale) that make up this variation.

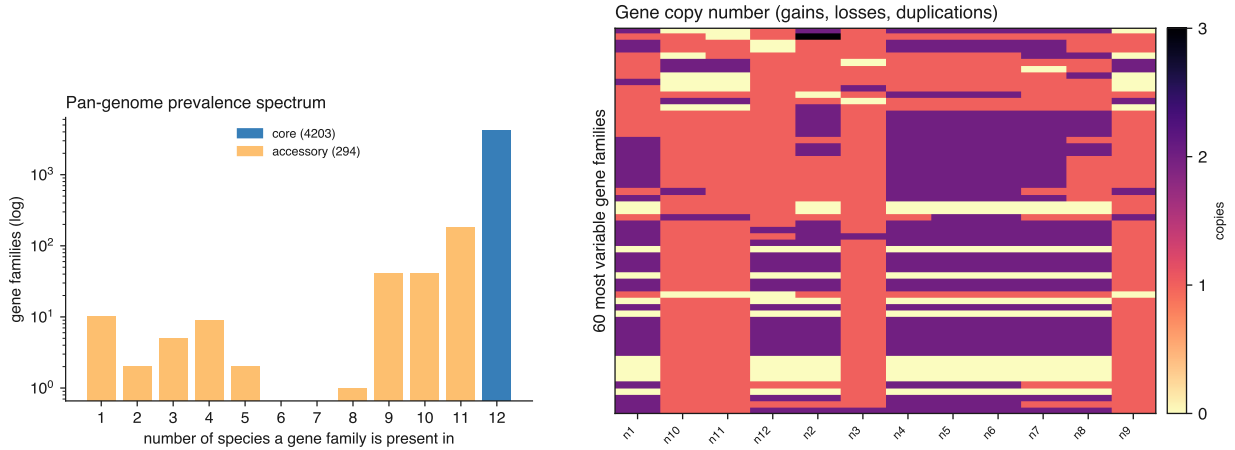


Figure 3. Left: pan-genome prevalence spectrum (log scale) — the U-shaped core/accessory split. **Right:** copy number of the 60 most variable gene families across the 12 genomes.

2.3 The events behind it, and gene-tree discordance

The whole history is one event log (Fig. 4, left): 788 losses, 330 pseudogenisations, 474 duplications, 496 transfers, 134 inversions, 47 transpositions and 18 originations. Because the model re-mints lineages at every event, ZOMBI2 reconstructs a gene tree for each gene atom. Duplications and transfers make these gene trees disagree with the species tree: Fig. 4 contrasts the species tree with the reconstructed tree of gene **b2876**, whose duplication history gives it several paralogous copies per lineage — a topology no strictly-vertical model could produce.



Figure 4. Left: structural events fired over the whole tree. **Right:** the species tree beside the reconstructed gene tree of **b2876** — paralogs from duplication make the gene tree discordant.

3 Summary

Starting from nothing but a GFF file and a tree, ZOMBI2's nucleotide genic model evolves a real 4,641,652 bp bacterial chromosome and delivers, in one self-consistent run: extant genomes with realistic sizes, a core/accessory pan-genome, gene duplications and transfers, pseudogenisation, and a ground-truth gene tree for every gene and intergene. Every gene stays intact throughout, so the output can be read as a known-truth benchmark for pan-genome, orthology and gene-tree/species-tree reconciliation methods on genomes with authentic architecture. The full run takes under a second.

Reproduce with `python run_analysis.py` in `ecoli_nt_analysis/` (tree seed 42, simulation seed 7). Generated with `ZOMBI2 zombi2 genomes -rate-model nucleotide -gff`.